

Abdelrahman Taeha

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EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science - Artificial Intelligence Concentration

Atlanta, GA

Aug. 2026 – May 2028 (Expected)

Sacramento State University

Bachelor of Science in Computer Science

Sacramento, CA

Aug. 2023 – Dec. 2025

EXPERIENCE

AI Research Student

Independent Research

August 2025 – Present

Remote

- Evaluated medical LLM agent integrity across 840 paired-twin scenarios for the MedInsider FHIR-style benchmark under institutional pressure
- Built evaluation and logging infrastructure, including deterministic action tracking and auditing workflows for 9 distinct medical vision-language models across full-pipeline and baseline sub-panels
- Trained a neutral baseline model and developed data collection pipelines for 2 major open-access clinical repositories utilized to analyze demographic bias in spinal radiology VLM outputs
- Co-authored SpineFairBench, a counterfactual benchmark evaluating 7,996 paired spinal radiographs to audit demographic sensitivity in VLM-based report generation

Software Engineer

DHCS

May 2025 – Present

Sacramento, CA

- Automated monitoring and reporting pipelines across enterprise systems, reducing manual observability overhead by 40% and cutting the mean time to detection (MTTD) for production incidents in half
- Engineered backend integrations and validation workflows that surfaced system behavior anomalies early, accelerating root-cause analysis by 35% and streamlining stakeholder communication
- Established internal dashboards and reporting applications to visualize agent-driven operational metrics, improving decision-making and reducing incident resolution time by 30%
- Implemented log-processing pipelines and monitoring integrations that enhanced system observability and improved real-time alerting accuracy by 20%

Software Engineering Intern (Machine Learning)

District Hut

May 2024 – Jul. 2025

Sacramento, CA

- Developed large-scale data pipelines processing 3TB/week across distributed cloud infrastructure, improving throughput and system reliability in a high-volume production environment
- Optimized data workflows and query performance, identifying and resolving bottlenecks that cut average pipeline latency and improved system stability under load
- Revamped data integrity checks with SQL, preventing five critical production issues related to data corruption, and automating data validation tasks by 75%, all while achieving 100% test coverage
- Collaborated with a team of 3 engineers to automate model retraining agents, reducing collective manual effort by 6 hours per week by triggering cloud-based retraining based on autonomous performance metrics

PROJECTS

Network Intrusion Detection System | Python, PyTorch, Flask, Redis, NumPy

- Teamed with 2 peers to develop an AI agent reasoning engine for threat detection, reaching 89% accuracy on NSL-KDD (125K+ samples)
- Engineered a decision-making layer using 41 traffic-flow features (protocol stats, TCP flag distributions, temporal patterns) to autonomously classify and flag malicious activity
- Architected a low-latency inference agent using Flask + Redis + Plotly, processing 50-100 network connections/sec with an interactive dashboard for “Agent-in-the-loop” monitoring

Al-Manar — Islamic Companion App (iOS) | Flutter, Dart, Provider, Google Places API, Xcode Cloud

- Led end-to-end development of Al-Manar, a production iOS application built with Flutter and Dart, managing the full software lifecycle from UI/UX design through App Store distribution via Xcode Cloud CI/CD
- Constructed a custom background audio framework extending BaseAudioHandler to enable seamless Quran recitation with system-level media controls, persistent playback state, and optimized memory handling for extended sessions
- Integrated AI-driven location agents using the Google Places API and the Haversine formula to autonomously calculate high-precision, GPS-based prayer times and real-time Masjid discovery for a global user base

TECHNICAL SKILLS

Programming: Python, Java, C/C++, SQL, JavaScript, Swift, MATLAB, R, COBOL

Frameworks/Platforms: React.js, Node.js, Spring Boot, AWS, Docker, Git, GCP

Data & ML: TensorFlow, scikit-learn, pandas, NumPy, SAP (Data Integration), model evaluation

Security/DevOps: SIEM, GitHub Actions, CI/CD, monitoring pipelines

AI & Agents: Autonomous Agents, LLM Orchestration (LangChain/CrewAI), RAG, Prompt Engineering